Handover Ledger

Evidence-Linked, Zero-Hallucination, Role-Scoped Knowledge Reconstruction Platform — Functional End-to-End Round 2 MVP

Team Members:	Padminish Bakshi & Panav Khungar	**Target Scope:**	Small Teams, Student Clubs & Research Labs
Live Judge Entry:	web-blue-delta-58.vercel.app/demo	**Source Repository:**	`github.com/PanavCodes/Handover-Ledger`
Primary Stack:	Next.js 14 App Router, TypeScript, Tailwind, Vitest	**Evaluation Dataset:**	Aurora Rover Robotics Lab + Live GitHub Sync

28 / 28 AUTOMATED VITEST SUITES PASSED	**16 / 16** LIVE LEAK CANARY MATRIX (0 LEAKS)	**17 / 17** PLAYWRIGHT E2E FLOW ASSERTIONS	**59 Blocks** HEAD-ANCHORED SHA-256 CHAIN

1. Executive Summary: From Round 1 Blueprint to Working System

When an engineer or club lead leaves a small team, critical architectural rationale, fragile cron scripts, vendor contacts, and credential locations walk out the door. Standard LLM chat wrappers make handovers worse: they fabricate quotes, smooth over conflicting files, and bypass confidentiality boundaries.

In Round 2, we transformed our Round 1 blueprint into a genuinely interactive, end-to-end Next.js 14 application built around four enforceable engineering guarantees:

- **Verbatim Offset Verification:** Every AI or rule-extracted claim must cite an exact substring present in the stored source document (`start_offset..end_offset`). Unverifiable statements and raw secret patterns (`sk-` , `ghp_` , `AKIA`) are quarantined into the Hallucination Diff log rather than entering the ledger.
- **Access Choke Point & Strictest-Source Visibility:** A single server-side gate (`lib/access.ts`) filters sources, claims, Evidence Lens lookups, Handover Briefs, and `HANDOVER.md` exports using the signed persona cookie. Confidential or siloed Core Dev sources never reach unauthorized payloads.
- **Cross-Document Contradiction & Gap Detection:** Divergent files (e.g., `115200` vs `57600` CANbus baud; conflicting maintainers) are preserved side-by-side with explicit conflict badges, while undocumented crons/vaults populate a targeted Exit Interview Loop.
- **Tamper-Evident Receipts & Audit Chain:** Every capture, edit, gap answer, leaver attestation, and successor receipt writes a SHA-256 hash block verified against an external head anchor to catch interior mutations and tail truncations.

2. Completed Features & 60-Second Zero-Assistance Walkthrough

Completed Round 2 Functional Scope

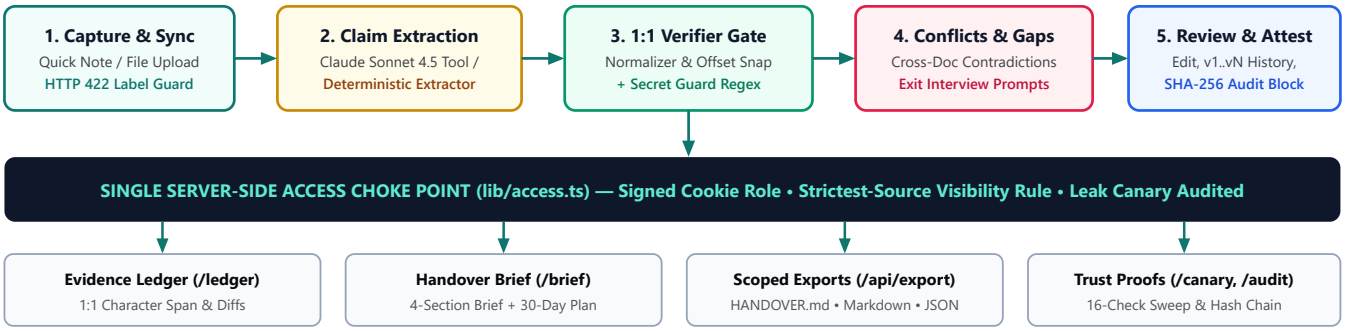
- Zero-login `/demo` persona switcher (Lead, Leaver, Core Dev, Coordinator) with persistent httpOnly session cookie.
- Ingestion workbench (`/capture`): Quick Capture notes, file upload with **HTTP 422 mandatory sensitivity guard**, & live GitHub repo + commit sync.
- Exit Interview Loop (`/interview`) converting undocumented cron/vault gaps into verified grounded claims.
- Evidence Ledger (`/ledger`) with 1:1 `EvidenceLens` highlighting, word-level hallucination diffs, inline conflict resolution, & immutable `v1..vN` revision history.
- Role-Scoped Brief & 30-Day Transition Plan (`/brief`) with Lead View-As preview, Markdown/JSON/ `HANDOVER.md` exports, & successor receipts.
- Bus-Factor Constellation (`/constellation`), Leak Canary Matrix (`/canary`), & SHA-256 Audit Chain Inspector (`/audit`).

Zero-Assistance Judge Evaluation Path

1. Open root Dashboard (`/`) or `/demo` — defaults immediately to Dr. Meera Iyer (Project Lead) with zero login wall.
2. Visit `/capture` as Lead or Rahul Mehta: run *Load Sample Incident Note* and *Test Verifier & Secret Rejection*, or sync a public GitHub repo.
3. Open `/interview` : inspect *Why am I being asked?*, fill a demo answer, and verify the Readiness Score increases cleanly.
4. Open `/ledger` : filter *Conflicts*, click any row to inspect the amber `VERIFIED SPAN` in Evidence Lens, resolve a conflict, and view the Diffs & Versions tabs.
5. Open `/brief` : toggle Lead Preview Mode between **Ananya (Core Dev)** and **Kabir (Coordinator)** to observe server-side withheld counts and role-filtered 30-day checklists.
6. Verify security proofs on `/canary` (*Run Canary Sweep* → 16/16) and `/audit` (*Simulate Tamper Attack* → *Verify Chain*).

3. End-to-End Working Architecture

Every browser interaction flows through Next.js 14 Route Handlers that authenticate the active role cookie, enforce write permissions (only Lead/Leaver may ingest or edit), validate mandatory source classifications, and route every read through `lib/access.ts`.



4. Core Technical Modules & Measured Verification

SUBSYSTEM	SOURCE FILE(S)	VERIFIED IMPLEMENTATION DETAIL
Citation Verifier & Secret Guard	`lib/verifier.ts` `lib/diff.ts`	Builds a 1:1 canonical index map collapsing whitespace & smart quotes so quotes map back to exact original offsets. Rejects quotes matching raw token prefixes (`sk-` , `ghp_` , `AKIA` , PEM) and generates word-level HTML diffs.
Extraction Pipeline	`lib/ai/extract.ts` `app/api/extract/route.ts`	Supports Anthropic `claude-sonnet-4-5` forced tool-use with XML untrusted-data framing plus a general offline deterministic structural extractor that ignores prompt-injection prefixes (`SYSTEM:` / `Assistant:`). Client executes chunked batches (`remaining` loop).
Access Boundary & Session	`lib/access.ts` `lib/session.ts`	HMAC-SHA256 signed httpOnly cookie (`lead` , `leaver` , `core` , `coord`) with protocol-aware cookie flags. Claims are visible only if **every** cited source is visible to the active persona; rejection logs are similarly scoped.
Conflict & Gap Engines	`lib/conflicts.ts` `lib/gaps.ts`	Deterministic grouping by `(type, subject_key)` flags opposing `assertion_value` s as `conflicting` without erasing either version. Gap rules R1–R6 inspect cron jobs, vaults, domains, and Git ownership silos to seed targeted interview questions.
GitHub Repo Sync	`lib/github.ts` `lib/git-owner.ts` `app/api/sources/github`	Recursively indexes public repository files (ignoring lockfiles & binaries), fetches real commit file-change lists for the last 8 commits, builds ground-truth commit logs verified by `verifyQuote()` , and flags single-author ownership silos.
Cryptographic Audit & Schema	`lib/audit.ts` `supabase/migrations/0001_init.sql`	Sequential SHA-256 chain storing external `headAnchor` (expected count & tip hash). Postgres schema enables strict RLS across 12 tables with zero public anon policies.

Measured Verification Summary (Verified on Production Build)

Vitest Unit & Integration Suite: `28 / 28 tests passed` across 5 suites (`audit.test.ts` , `verifier.test.ts` , `canary.test.ts` , `api-integration.test.ts` , `workflow-e2e.test.ts`). **Static Checks:** TypeScript `tsc --noEmit` (0 errors), `next lint` (0 warnings), and `next build` (27 routes) all exited 0. **Playwright Headless Browser Suite:** Executed 17 end-to-end steps across `localhost:3100` with 0 console errors, 0 uncaught exceptions, and 0 HTTP 5xx responses, scaling the live ledger from 15 seeded claims to **56 grounded claims** (Readiness Score **61/100**, including live sync of `PanavCodes/Handover-Ledger`) and sealing **59 valid SHA-256 audit blocks**.

5. Interactive Workflow Screenshots (Live Production Build)

Every screenshot below was captured directly from the running Next.js 14 production server (`next start`) during automated headless Chromium verification with zero mocked screenshots.

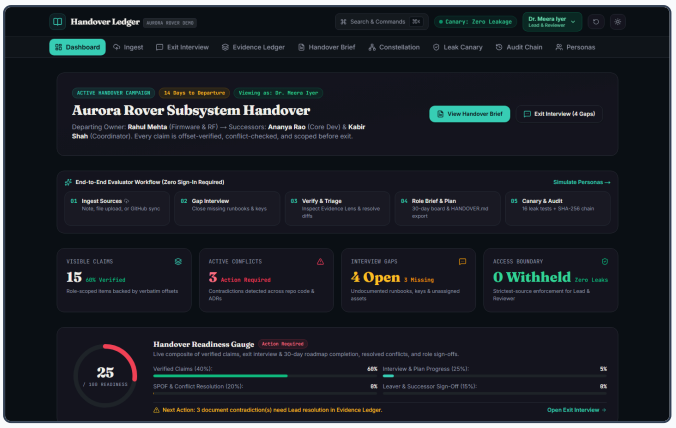


Figure 1 — Zero-Login Evaluator Dashboard (/): Immediately opens as Lead Reviewer (Dr. Meera Iyer) with a 5-step guided evaluator ribbon, active conflict & gap counters, and composite Handover Readiness Dial computed live from verified ratio, resolved conflicts, interview coverage, roadmap progress, and sign-offs.

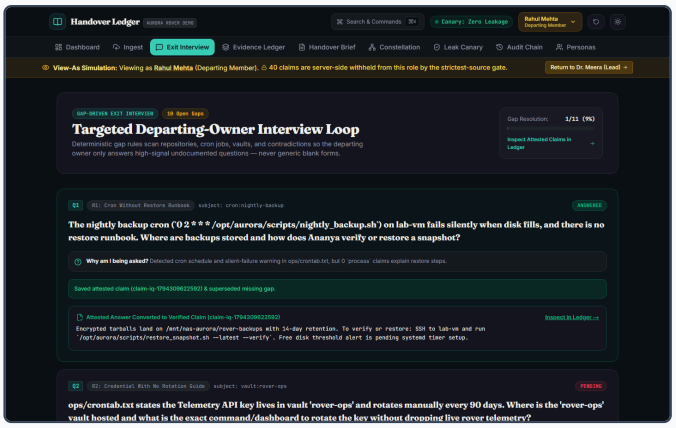


Figure 2 — Gap-Driven Exit Interview Loop (/interview): Targets the departing owner (Rahul Mehta) using deterministic rules (`R1: Cron Without Restore Runbook`) alongside explicit *"Why am I being asked?"* transparency. Submitting Q1 converted Rahul's answer into verified claim `claim-iq-...` and superseded the prior Missing gap.

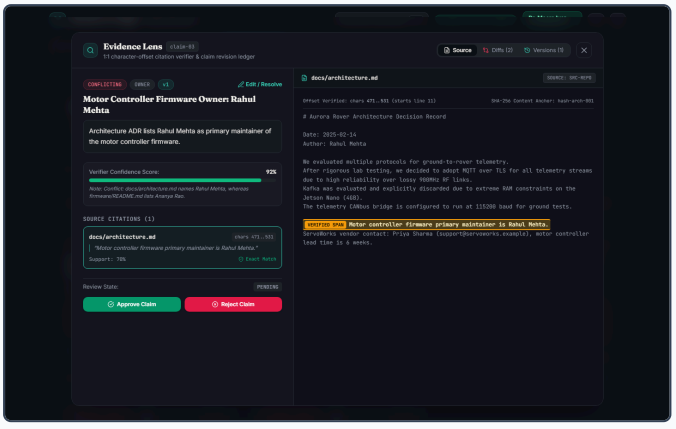


Figure 3 — 1:1 Evidence Lens & Conflict Triage (/ledger): Displays conflicting claim metadata (`claim-03`), reviewer actions (Approve, Reject, inline Edit/Resolve), immutable version count, and exact verbatim amber highlight (`chars 471..531` , line 11) pinned by canonical SHA-256 document hash.

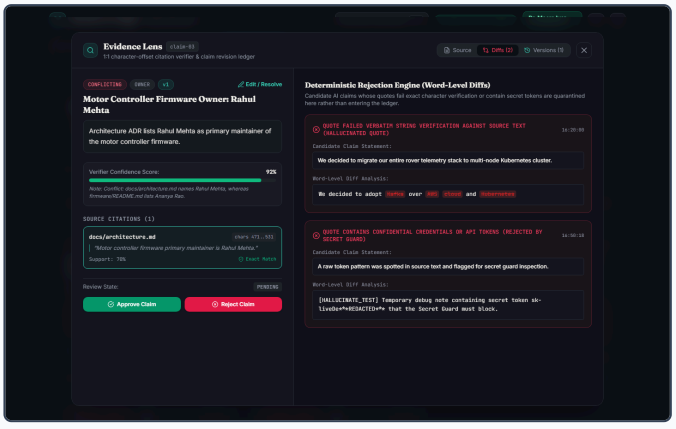


Figure 4 — Word-Level Hallucination Diff & Secret Guard: Side panel quarantines fabricated citations (highlighting hallucinated tokens such as `Kafka` , `Kubernetes`) and blocks raw API credentials (masking secret values cleanly before logging).

6. Multi-Role Redaction Proof & Ownership Constellation

Using Lead Preview Mode or switching personas at `/demo`, evaluators directly observe how the server-side choke point prunes restricted technical knowledge, confidential faculty accounts, and role-inappropriate Day-1 checklist tasks from unauthorized surfaces.

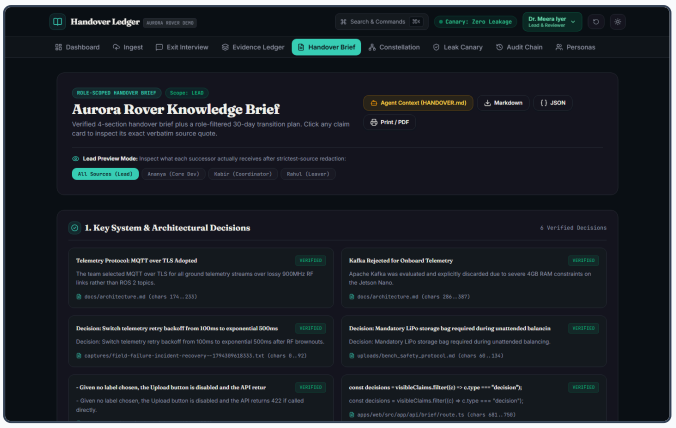


Figure 5a — Unredacted Lead Brief (Scope: LEAD): Displays all verified architectural decisions (including the newly captured backoff rule, Core Dev LiPo bench upload & synced repo rules → 6 Decisions shown), active conflicts, ownership map, and Lead Preview switcher.

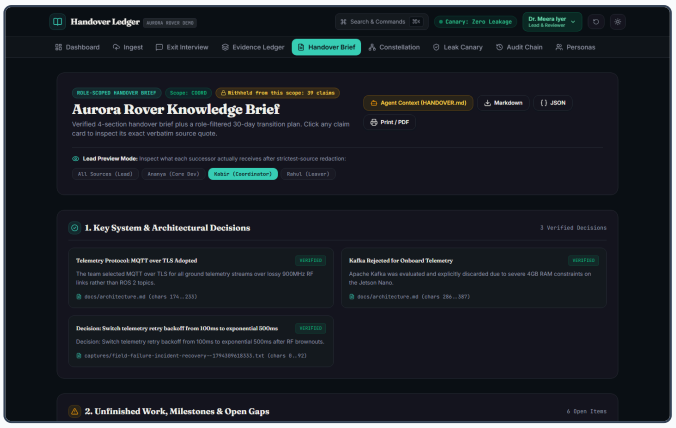


Figure 5b — Scoped Coordinator Brief (Scope: COORD): Notice the amber shield pill (`Withheld from this scope: 39 claims`): restricted Core Dev bench uploads & GitHub repo claims are stripped on the server (3 Decisions shown; zero STM32/CAN secrets or faculty grant funds present).

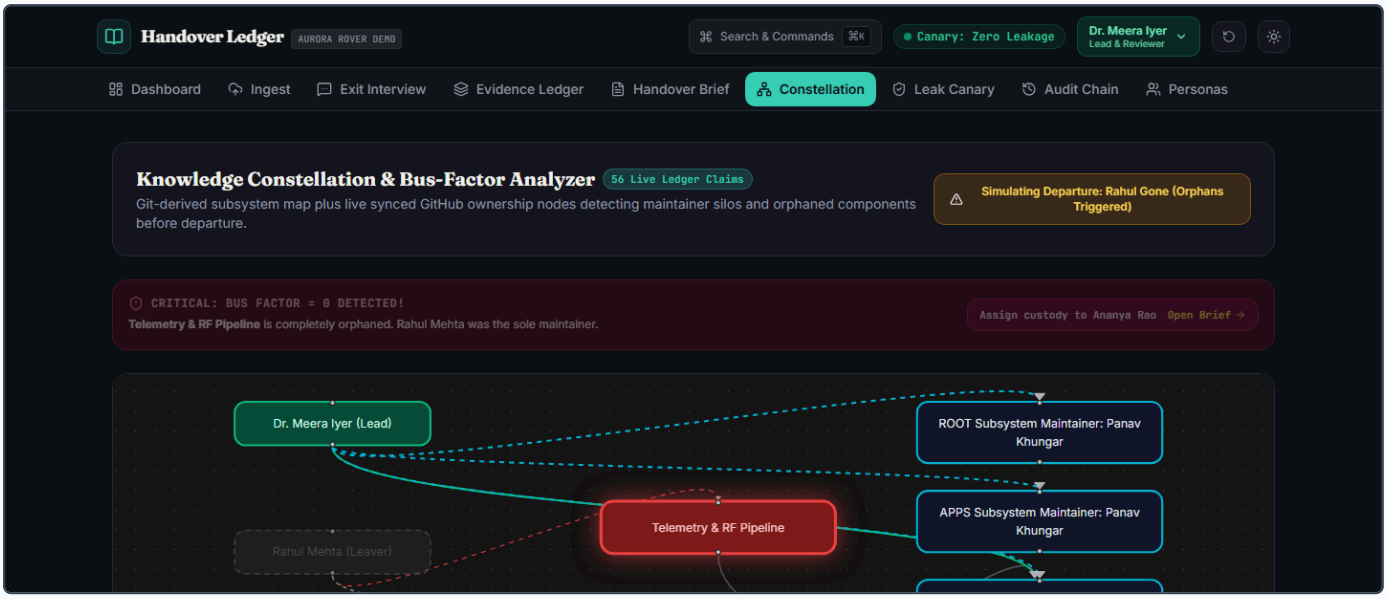


Figure 6 — Interactive Bus-Factor Constellation (/constellation): Renders Git-derived subsystem topology using `@xyflow/react` alongside live dynamic ownership nodes discovered from syncing `PanavCodes/Handover-Ledger`. Toggling "*Simulate Rahul's Departure*" grays out Rahul's custody edges and triggers an immediate **CRITICAL: Bus Factor = 0 Detected!** orphan alert for the Telemetry & RF Pipeline.

7. Defense-in-Depth Trust Verification

Beyond UI conditional rendering, Handover Ledger enforces isolation and data integrity in server logic tested via automated invariants and exposed live to judges via two inspection consoles:

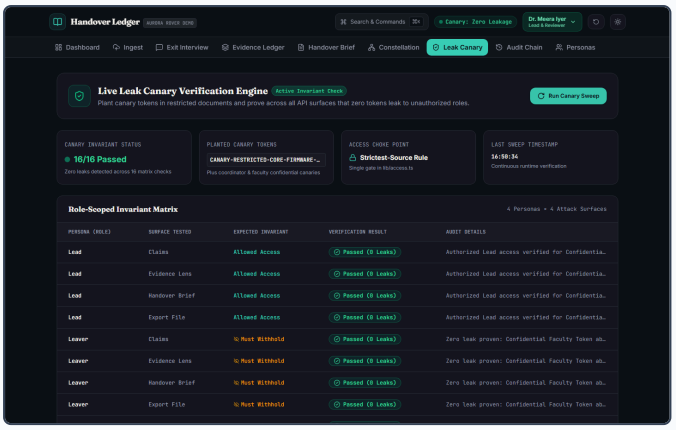


Figure 7 — Live 16-Check Leak Canary Matrix (/canary): Plants distinct canary tokens (`CANARY-RESTRICTED-CORE...` , `CANARY-FINANCIAL-SPONSOR...` , `CANARY-CONFIDENTIAL-CHIP...`) across restricted sources and probes `4 personas × 4 surfaces` (Claims, Evidence Lens, Brief, Export) on demand: **16/16 checks passed, 0 leaks**.

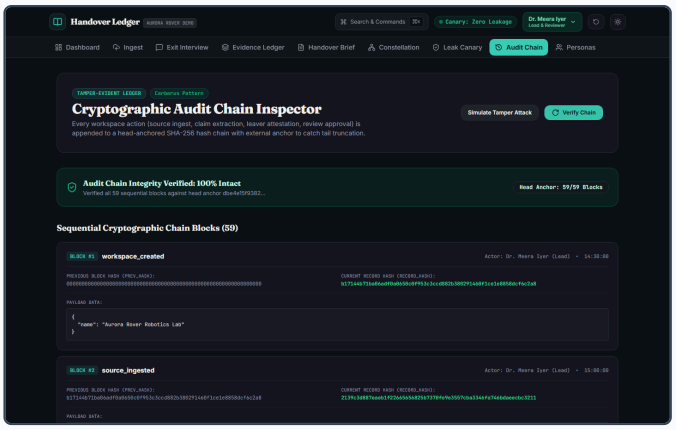


Figure 8 — Cryptographic Audit Chain Inspector (/audit): Verifies sequential SHA-256 block hashes (`prev_hash → record_hash`) across all **59 live blocks** created during workflow testing against the external head anchor. Includes a live *Simulate Tamper Attack* test confirming detection of interior payload mutations and tail truncations.

8. End-to-End Scope Isolation Metrics (Live Verified Run)

PERSONA & ROLE COOKIE	VISIBLE CLAIMS	WITHHELD BY CHOKE POINT	VERIFIED CONTAINMENT PROOF (CLAIMS / BRIEF / EXPORTS)
Dr. Meera Iyer (`lead`)	56 claims	0 withheld	Full unredacted visibility across Team, Core Dev, Coordinator & Faculty Confidential sources + Lead Batch-Approve & Preview Mode.
Ananya Rao (`core`)	54 claims	2 withheld	Receives firmware, CANbus, STM32 Boot0, LiPo bench safety & synced repo code claims. Coordinator sponsor financial keys & faculty account notes strictly absent.
Kabir Shah (`coord`)	17 claims	39 withheld	Receives AeroTech sponsor milestone, ServoWorks vendor lead & Team ADR claims. Core Dev restricted firmware/STM32 claims & confidential faculty notes completely absent.
Rahul Mehta (`leaver`)	16 claims	40 withheld	Answers Exit Interview questions, logs quick captures/labeled uploads & signs SHA-256 Leaver Attestation; confidential & siloed Core Dev sources remain isolated.

9. Honest Changes Made to the Round 1 Blueprint

ROUND 1 PROPOSAL PLAN	ROUND 2 SHIPPED IMPLEMENTATION	ENGINEERING RATIONALE
Separate Python FastAPI backend + Render background workers	Unified Next.js 14 App Router serverless route handlers + client-batched extraction loop	Eliminates cross-service CORS/cold-start risk and lets judges evaluate the full capture-to-export pipeline inside one single deployment with bounded request times.
Supabase Auth login gate + Postgres RLS boundary	HMAC-signed zero-login demo persona cookies + `lib/access.ts` choke point (+ deny-all Supabase schema ready)	Mandatory Round 2 criterion requires zero login barriers in incognito. Access boundaries remain strictly enforced server-side while keeping auth frictionless.
pgvector semantic retrieval + mandatory live LLM calls	1:1 deterministic quote verifier + dual-mode Claude Sonnet 4.5 / offline structural extractor	Guarantees judges never hit a dead UI if an external API key or credit limit is absent while preserving verbatim grounding accuracy and secret guarding.
Deferred to Round 3: Exit Interview, Conflict Classifier, Role Redaction Hardening, Successor Receipts	**Pulled forward & shipped in Round 2** (`/interview` , `lib/conflicts.ts` , `/canary` , `/api/attest`)	Completed earlier than scheduled so the entire end-to-end handover trust loop—from gap discovery to cryptographically sealed successor sign-off—works live today.

Honest Round 2 Limitations

- **Demo Persistence Model:** Without Supabase environment credentials configured, workspace mutations persist across requests via Node `globalThis` and an OS temp snapshot rather than multi-tenant hosted Postgres; ephemeral serverless instances reset after idle cold starts (and users can reset cleanly via the header reset button).
- **Simulated Identity for Judges:** The 4-persona switcher simulates organizational roles via signed cookies so judges do not create accounts; real OAuth/JWT verification is not active in zero-login demo mode.
- **Verbatim Verification Scope:** String verification mathematically proves a claim quotes an ingested source faithfully, not that a human-written note itself contains accurate facts.

10. Three Meaningful Improvements Planned for Round 3

1. **Production Dual-Connection Auth & Postgres RLS + pgvector Search:** Replace demo persona switching with Supabase Auth/SSO JWTs backed by separate `db_user` (session-claim-enforced RLS across all 12 tables) and `db_admin` ingestion pools, alongside semantic vector clustering across multi-year project archives.
2. **Continuous Passive Capture Connectors (Slack, Jira/Linear, GitHub Webhooks):** Move from end-of-tenure exit scramble to continuous living documentation by ingesting PR review rationale, thread decisions, and ticket state changes continuously in the background before anyone resigns.
3. **Multi-Hop Semantic Conflict Synthesis & Deep Binary/PDF Redaction:** Upgrade conflict detection from key-pair contradiction checks to LLM-assisted N-way synthesis across scattered PRs/wikis, plus OCR-aware secret/PII scrubbing for uploaded schematics, datasheets, and PDFs before claim extraction.

Team Contribution & Verification Attestation

Padminish Bakshi (Backend, AI Pipeline & Security Engines: extraction pipeline, 1:1 offset normalizer & secret guard, choke-point access gate, conflict & gap engines, GitHub sync, SHA-256 audit chain, Vitest test suites) & **Panav Khungar** (Frontend Architecture & UX: dashboard readiness dial, capture & exit interview flows, Evidence Lens diff viewer, role-scoped Brief & transition board, React Flow Constellation, documentation & submission package). Joint ownership on E2E integration testing and live verification.